ITNE2005R

Network Engineering

Assignment ¾:

Site to Site VPN Configuration Using pfSense

Group Details:

Name

Student ID

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1. Introduction

A Site-to-Site VPN (Virtual Private Network) provides a secure communication channel between two geographically separated networks using the public internet. The purpose of this project is to configure an **IPsec VPN** between two remote office sites using **pfSense firewalls** to enable encrypted and private data transfer between local area networks (LANs).

Objectives:

- Establish a secure tunnel between two sites using IPsec VPN.
- Configure clients at each site to communicate securely.
- Demonstrate encrypted data transfer.
- Implement firewall rules and test connectivity.

2. Configuration Details

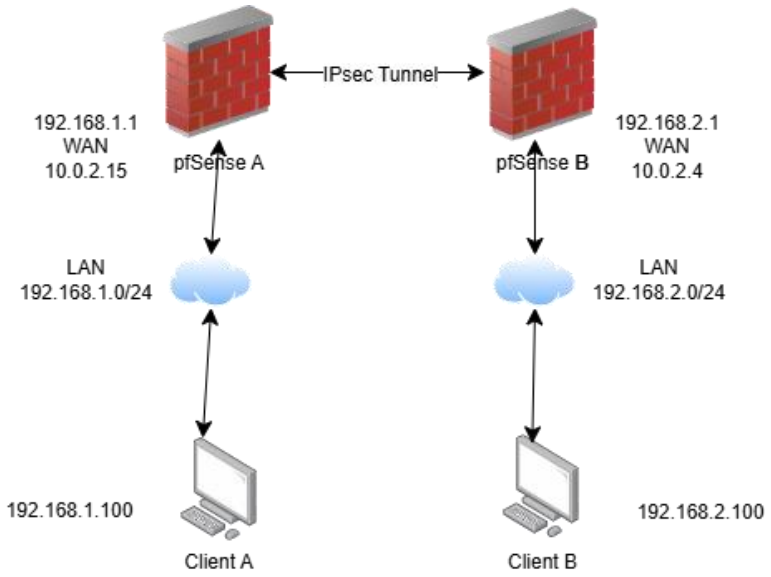
2.1 Tools and Environment

- **Software Used:**
  - Oracle VirtualBox
  - pfSense 2.6.0 (Community Edition)
  - Ubuntu 22.04 (as LAN clients)
- **Hardware:** Virtual Machines (2 pfSense, 2 Ubuntu)
- **VPN Type:** IPsec (IKEv2)
- **Encryption:** AES-256 / SHA-256 / DH Group 14

Figure 1

2.2 Network Setup

Site	WAN Interface	LAN Interface	Client IP	Description
Site A (pfSenseA)	10.0.2.15	192.168.1.1	192.168.1.100	Head Office
Site B (pfSenseB)	10.0.2.4	192.168.2.1	192.168.2.100	Branch Office



- **pfSense A** acts as the firewall for Site A, with interfaces:
  - WAN: 10.0.2.4/24 (connected to WAN-NET)
  - LAN: 192.168.1.1/24 (connected to LAN_A)
- **pfSense B** acts as the firewall for Site B, with interfaces:
  - WAN: 10.0.2.15/24 (connected to WAN-NET)
  - LAN: 192.168.2.1/24 (connected to LAN_B)
- Clients (A and B) each belong to their respective LANs and communicate through pfSense gateways.

2.3 pfSense Configuration (Step-by-Step)

Step 1: WAN/LAN Interface Setup

- Set **WAN** interface to DHCP.
- Assign **LAN** interface static IPs:
  - Site A: 192.168.1.1/24
  - Site B: 192.168.2.1/24

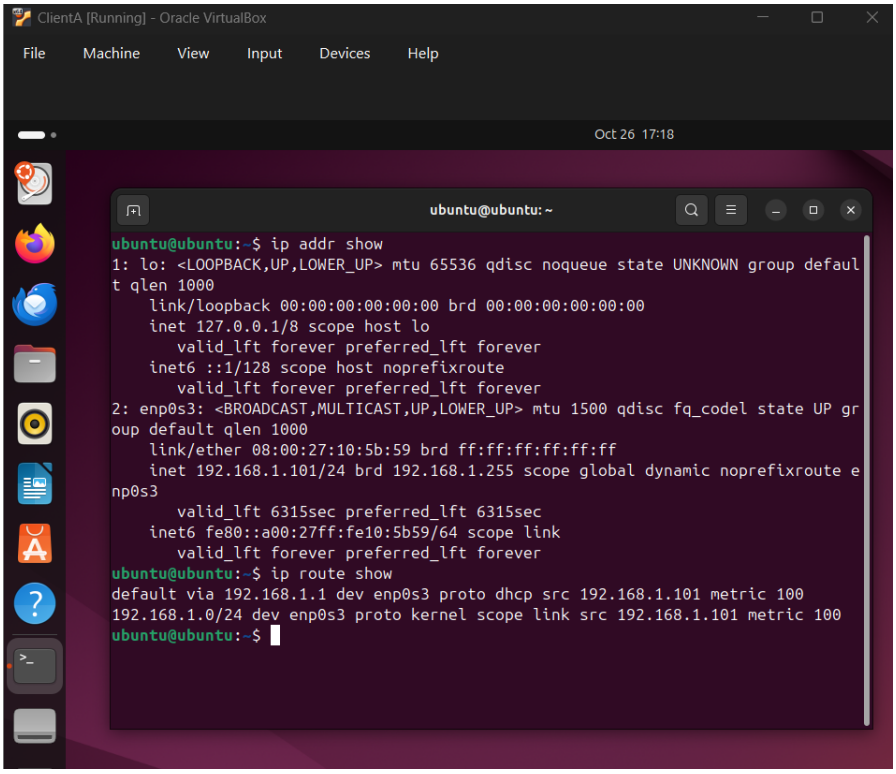


Figure 2

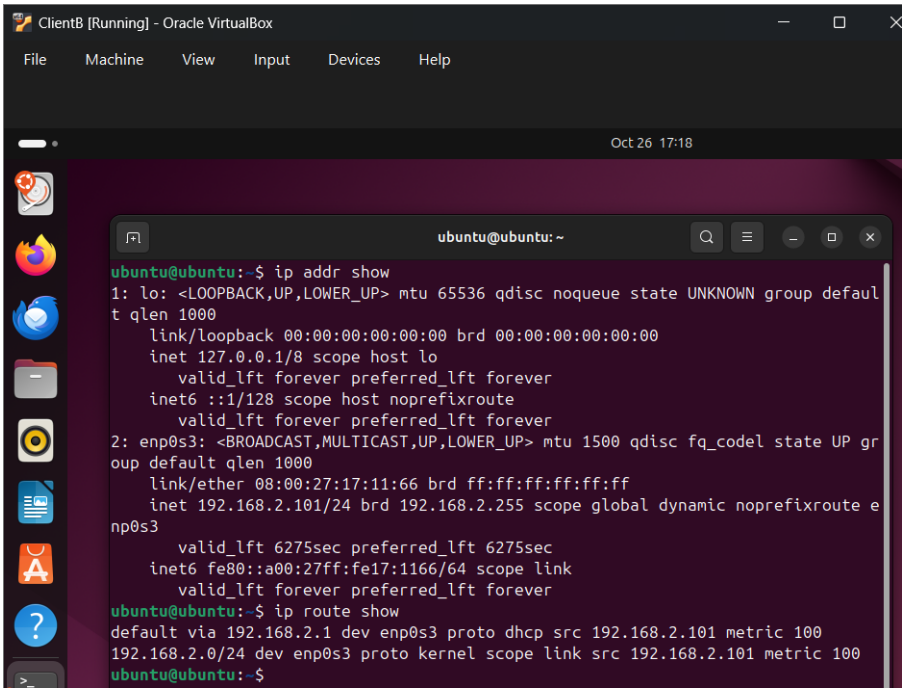


Figure 3

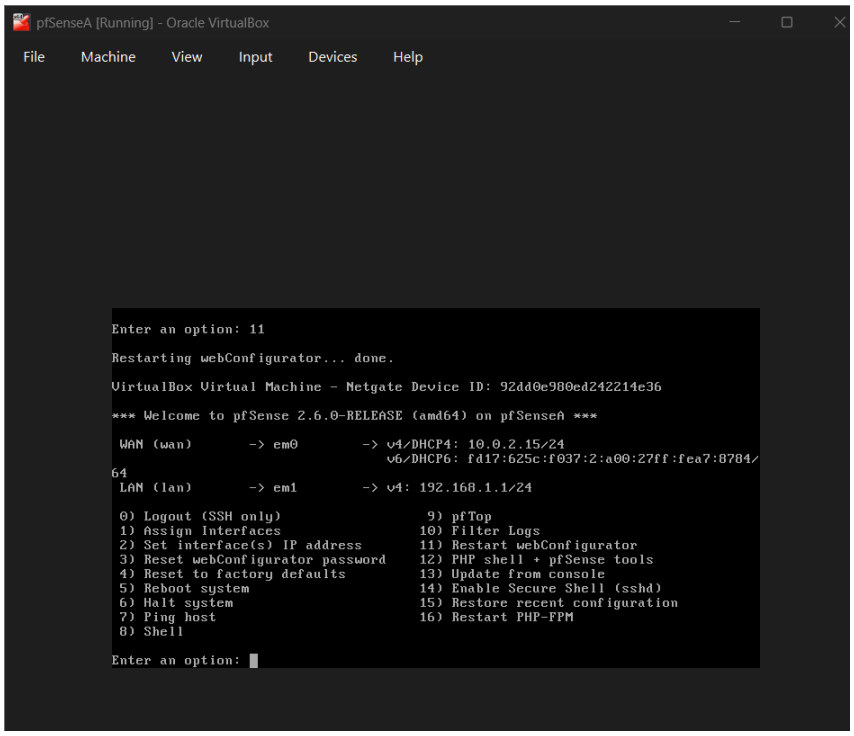


Figure 4

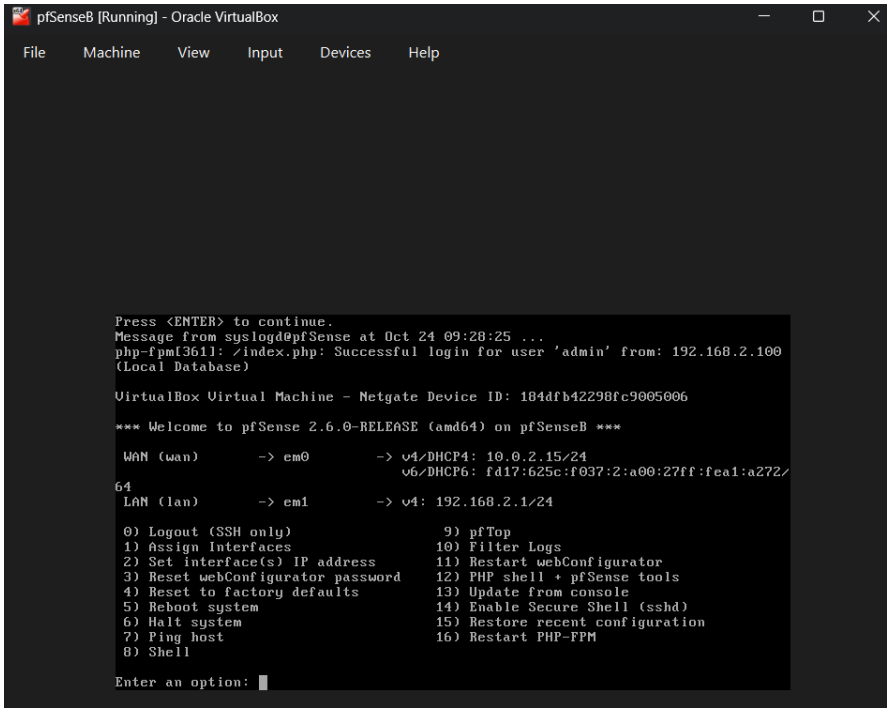
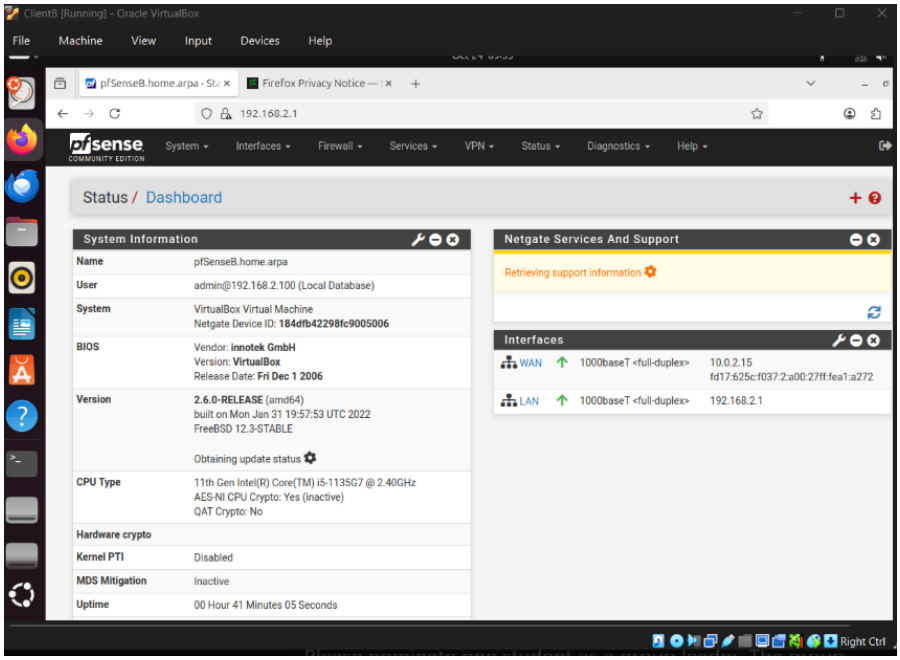


Figure 5



Step 2: IPsec Phase 1 Configuration

- Go to **VPN → IPsec → Add Phase 1**
- **Interface:** WAN
- **Remote Gateway:** (opposite site's WAN IP)
  - Site A → 10.0.2.4
  - Site B → 10.0.2.15
- **Authentication Method:** Mutual PSK
- **Pre-Shared Key:** `pfsense123`
- **Encryption Algorithm:** AES (256 bits)
- **Hash:** SHA256
- **DH Group:** 14 (2048-bit)
- **Lifetime:** 28800 seconds

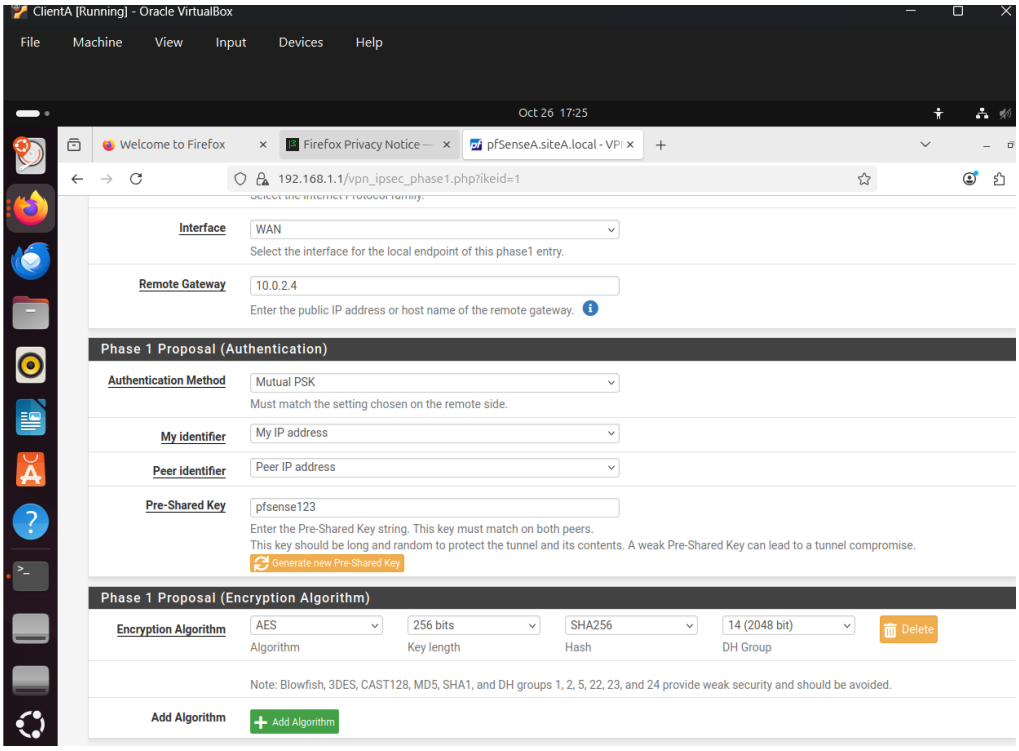


Figure 6

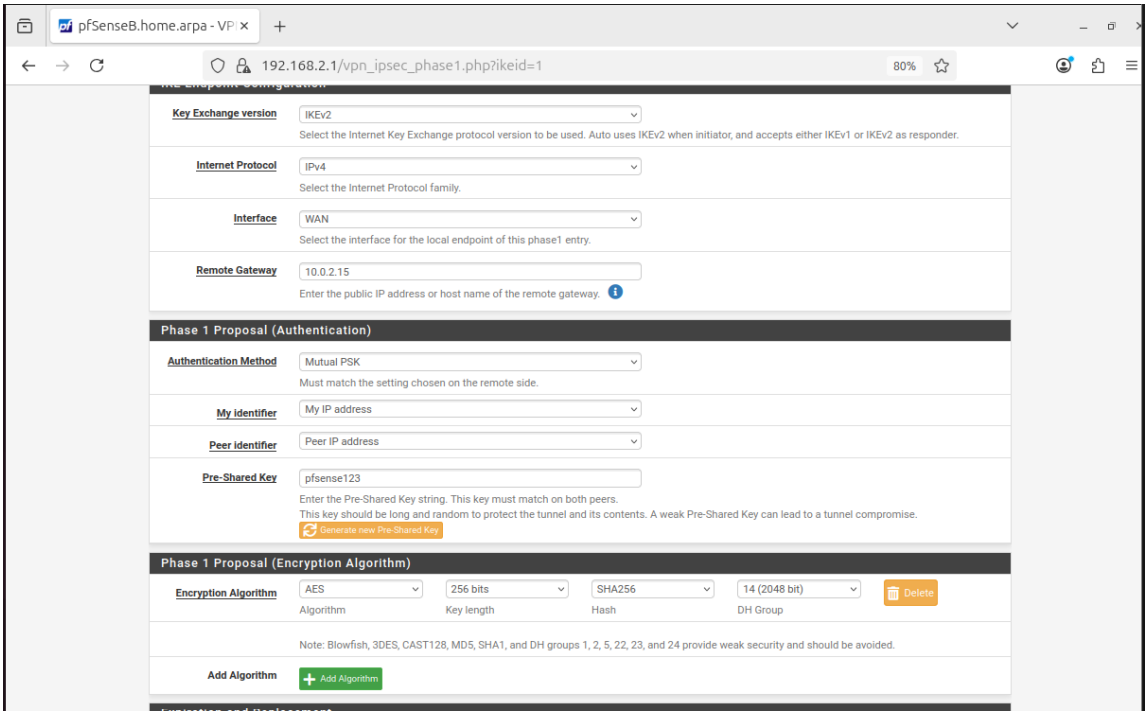


Figure 7

Step 3: IPsec Phase 2 Configuration

- **Local Network:** 192.168.1.0/24 (Site A)
- **Remote Network:** 192.168.2.0/24 (Site B)
- **Protocol:** ESP
- **Encryption:** AES-256, SHA256
- **PFS Key Group:** 14
- Save and apply.

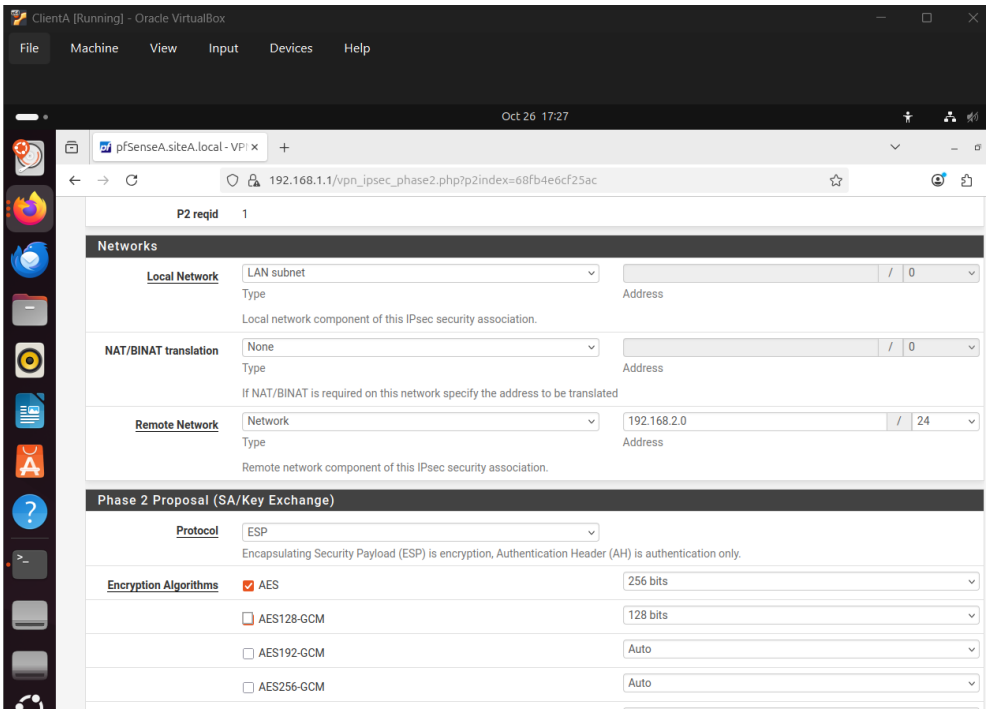


Figure 8

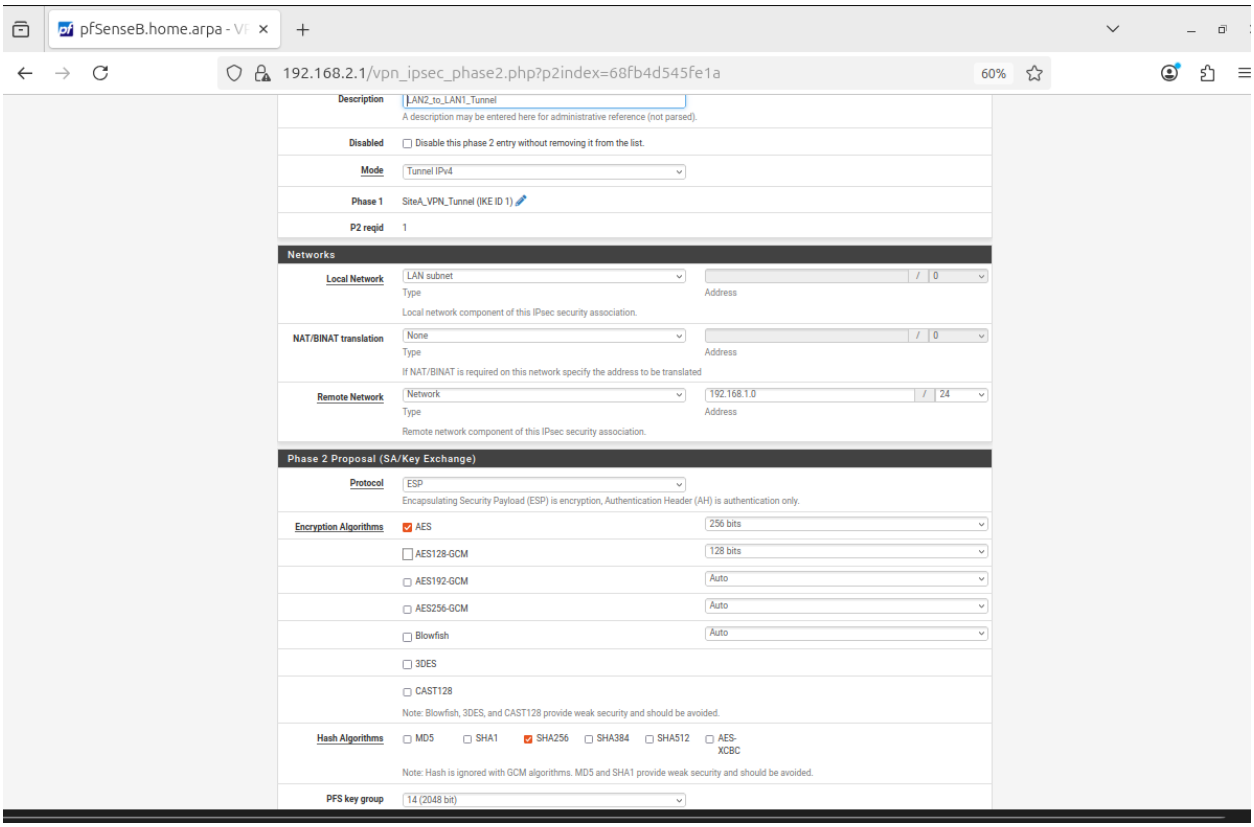


Figure 9

Step 4: Firewall Rules

- Go to **Firewall → Rules → IPsec**
  - Add rule: Action **Pass**, Protocol **Any**, Source **any**, Destination **any**
- Go to **Firewall → Rules → LAN**
  - Allow all traffic to pass to remote LAN.

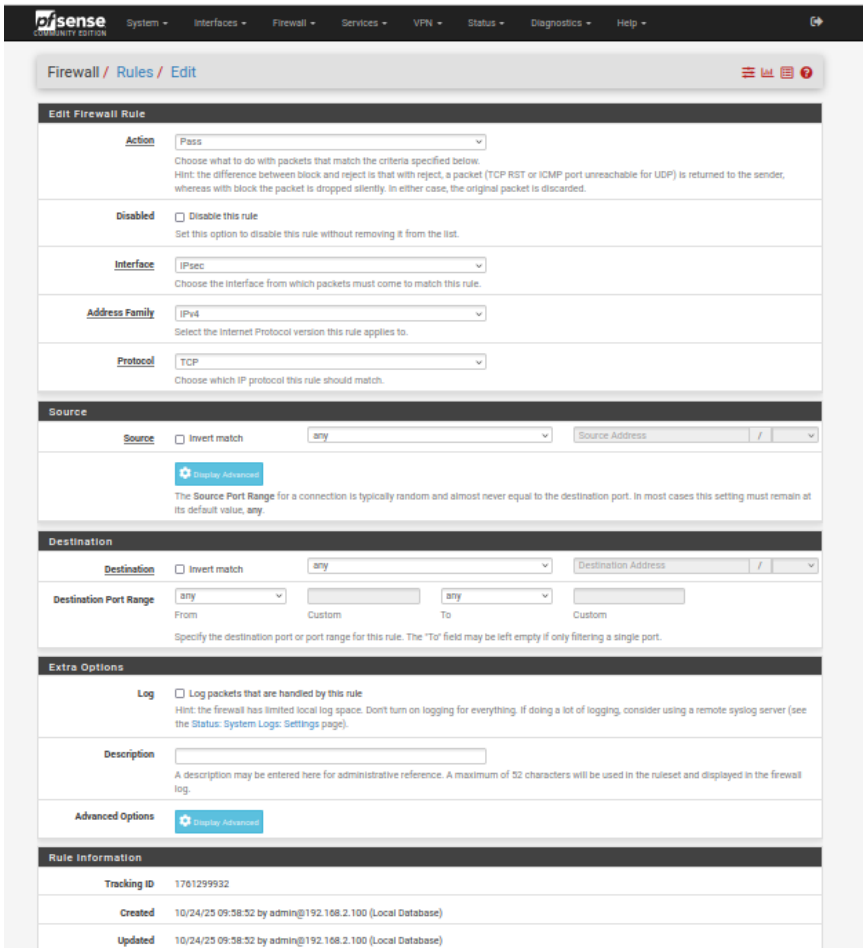

pfsense
COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
Firewall / Rules / Edit
Edit Firewall Rule
Action
Pass
Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.
Disabled
Disable this rule
Set this option to disable this rule without removing it from the list.
Interface
IPsec
Choose the interface from which packets must come to match this rule.
Address Family
IPv4
Select the Internet Protocol version this rule applies to.
Protocol
TCP
Choose which IP protocol this rule should match.
Source
Source
Invert match
any
Source Address
Display Advanced
The Source Port Range for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, any.
Destination
Destination
Invert match
any
Destination Address
Destination Port Range
any
From
Custom
any
To
Custom
Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.
Extra Options
Log
Log packets that are handled by this rule
Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the Status: System Logs: Settings page).
Description
A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset and displayed in the firewall log.
Advanced Options
Display Advanced
Rule Information
Tracking ID
1761299932
Created
10/24/25 09:58:52 by admin@192.168.2.100 (Local Database)
Updated
10/24/25 09:58:52 by admin@192.168.2.100 (Local Database)

Figure 10

Figure 11

Step 5: Tunnel Verification

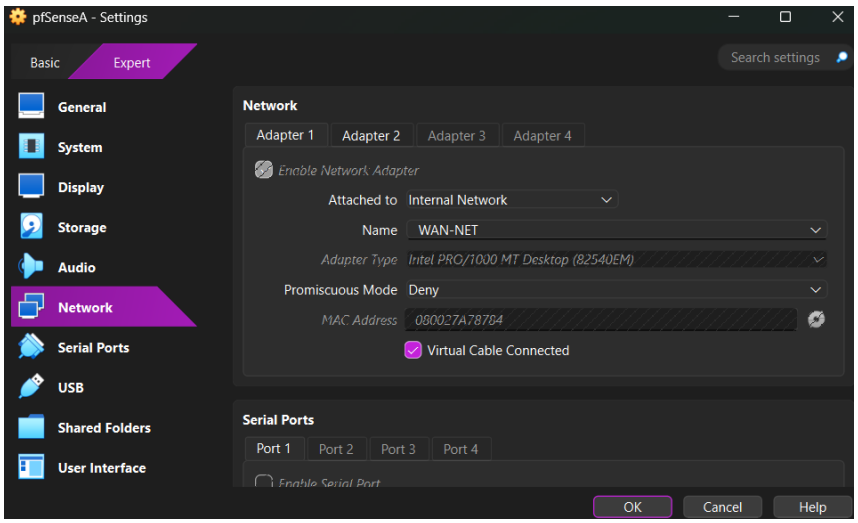
- Navigate to **Status → IPsec → Overview**
- Status should show:
  "**Established**" on both pfSenseA and pfSenseB.

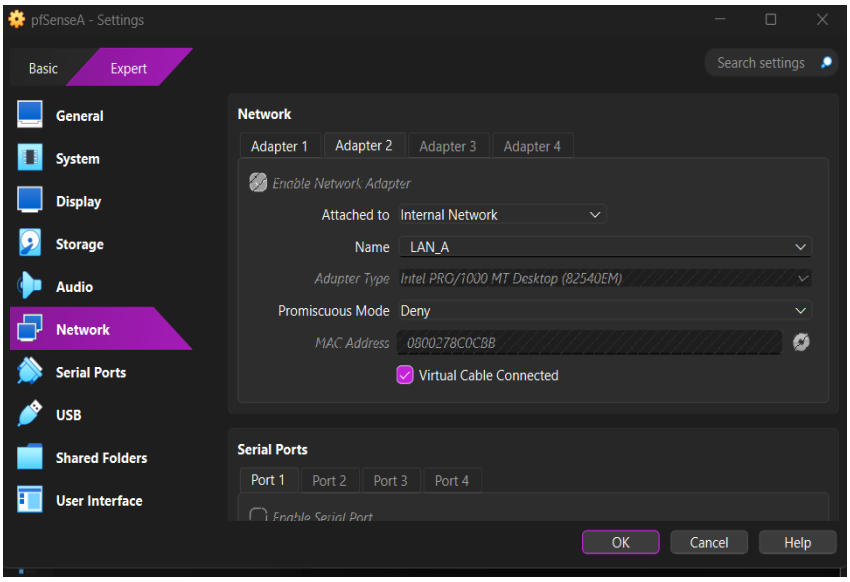
VirtualBox Adapter Configuration

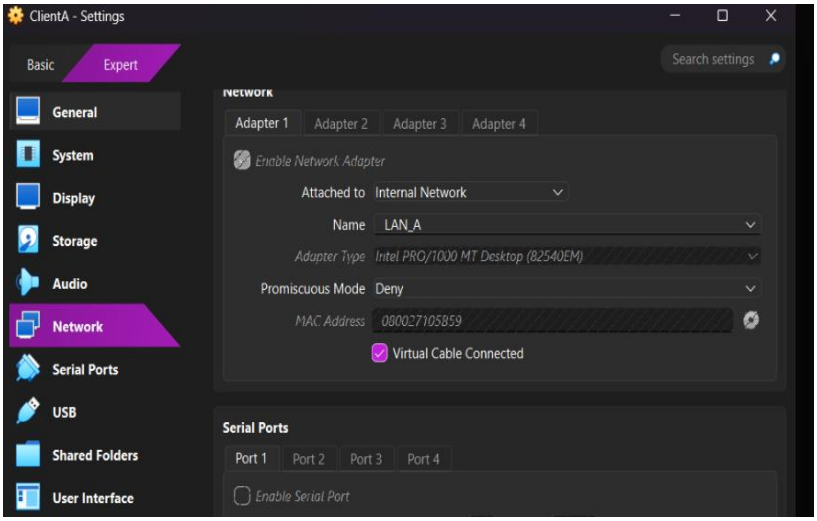
VM	Adapter	Attached To	Network Name	Function
pfSense A	Adapter 1	Internal Network	WAN-NET	Simulated Internet

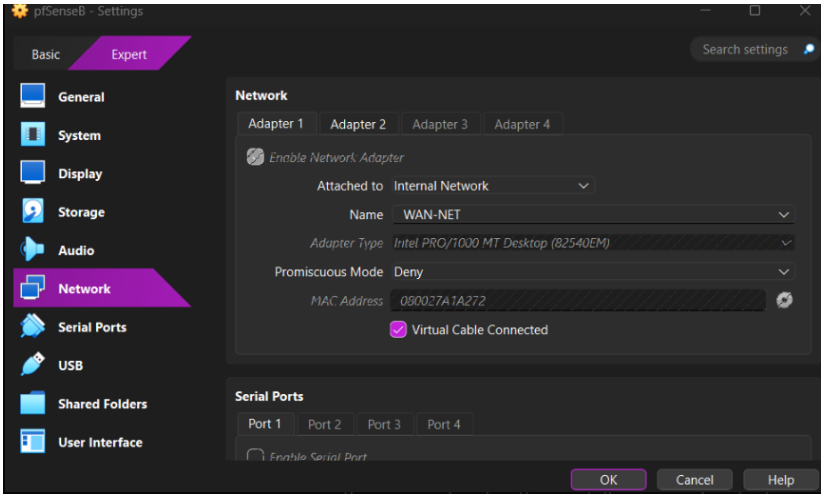
pfSense A	Adapter 2	Internal Network	LAN_A	Local network for Site A
Client A	Adapter 1	Internal Network	LAN_A	Client system in Site A
pfSense B	Adapter 1	Internal Network	WAN-NET	Simulated Internet
pfSense B	Adapter 2	Internal Network	LAN_B	Local network for Site B
Client B	Adapter 1	Internal Network	LAN_B	Client system in Site B

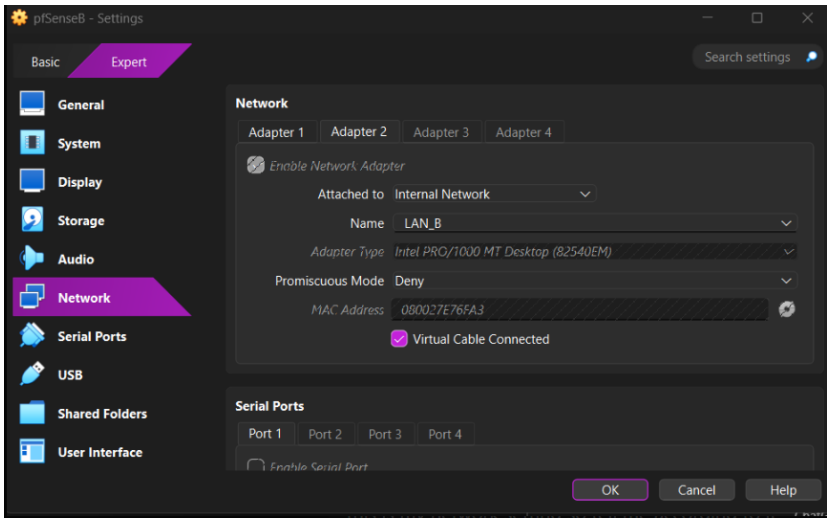
Explanation:
Each pfSense router connects one interface to the shared WAN-NET (for VPN communication) and another to its LAN. Clients are connected to their local pfSense instances via their LAN networks.

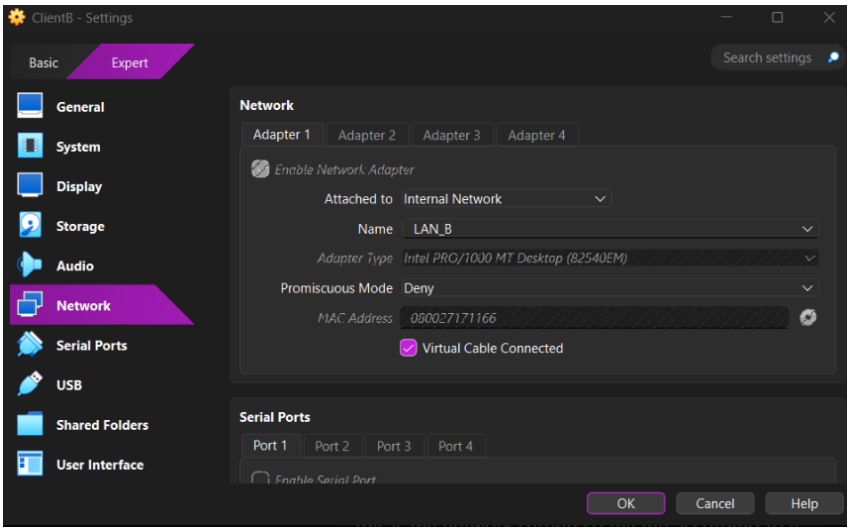





ClientA - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
Internal Network
Name
LAN_A
Adapter Type
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode
Deny
MAC Address
080027105B59
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port


pfSenseB - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
Internal Network
Name
WAN-NET
Adapter Type
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode
Deny
MAC Address
080027A1A272
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
OK
Cancel
Help


pfSenseB - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
Internal Network
Name
LAN_B
Adapter Type
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode
Deny
MAC Address
080027E76FA3
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
OK
Cancel
Help



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3. Security Measures

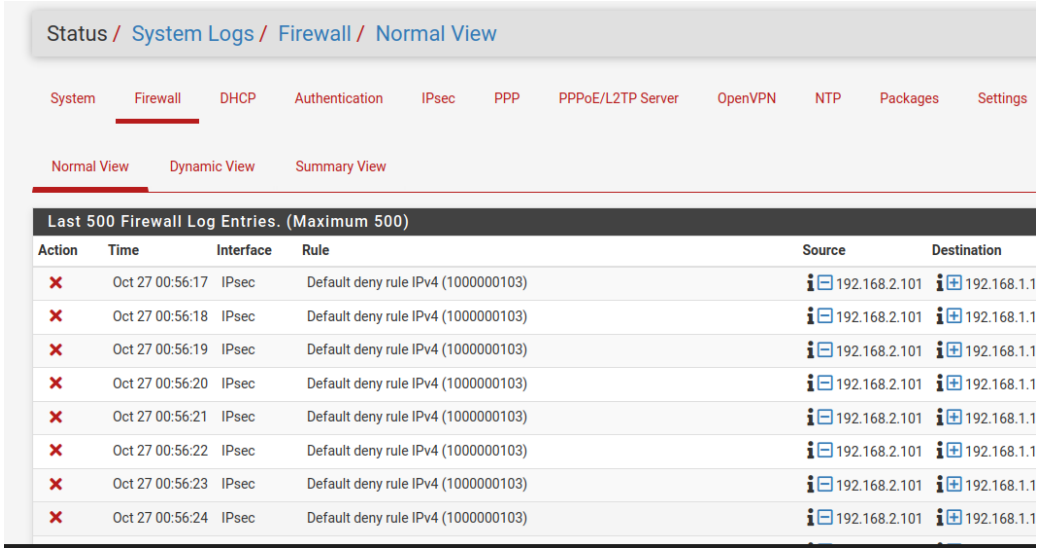
3.1 Encryption

The VPN uses AES-CBC (256-bit) for data encryption, HMAC-SHA2-256 for authentication, and MODP_2048 for key exchange. This ensures strong encryption and data confidentiality between both LANs.

- **Encryption Algorithm:** AES-256 ensures strong confidentiality.
- **Integrity Algorithm:** SHA-256 ensures data integrity.
- **DH Group 14:** Provides strong key exchange security.
- **IKEv2 Protocol:** Supports automatic rekeying and reliability.

3.2 Firewall Protection

- Only IPsec and LAN-to-LAN communication is permitted.
- Unwanted inbound WAN traffic is blocked by default.
- Firewall rules created under IPsec allow **only encrypted traffic**.



The pfSense firewall correctly blocked ICMP packets from 192.168.2.101 to 192.168.1.101, as shown in the IPsec interface deny logs. This confirms that firewall policies were successfully applied to secure communication across the VPN.

Data Transfer Demonstration

5.1 Ping Verification

From Client A:

```
ping 192.168.2.101
```

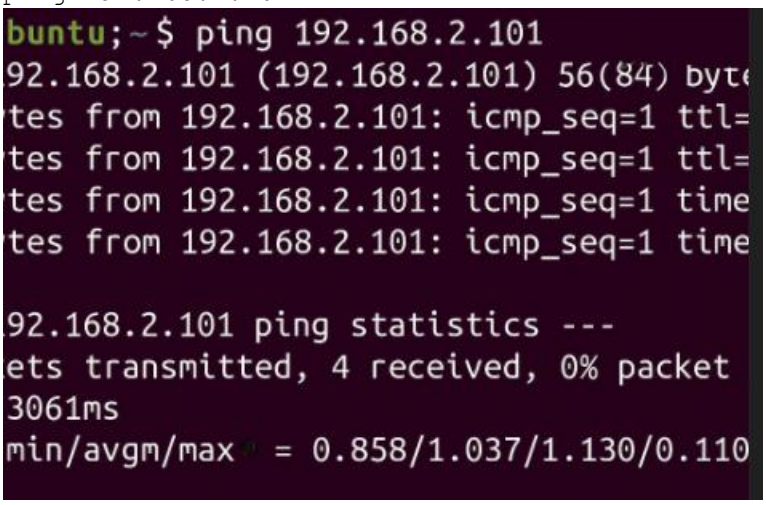


Figure 13

Result: Received successful encrypted data transfer.

From Client B:

```
ping 192.168.1.101
```

```
ubuntu@ubuntu:~$ ping 192.168.1.101
PING 192.168.1.101 (192.168.1.101) 5684) bytes of data.
64 bytes from 192.168.1.101: icmp_seq=1 ttl=64 time=0.505 ms
64 bytes from 192.168.1.101: icmp_seq=2 ttl=64 time=0.435 ms
64 bytes from 192.168.1.101: icmp_seq=3 ttl=64 time=0.507 ms
64 bytes from 192.168.1.101: icmp_seq=4 ttl=64 time=0.492 ms

---- 192.168.1.101 ping statistics ----
4 packets transmitted, 4 received, 0% packet loss, time 3002m

rtt min/avg/max/mdev = 0.435/0.484/0.507/0.032 ms
```

Figure 14

Result: Received two-way connectivity confirmed.

5.2 Tunnel Logs

- `Status ▸ IPsec ▸ Overview → Established`
- Firewall logs show ESP (protocol 50) traffic between 10.0.2.15 ↔ 10.0.2.4.

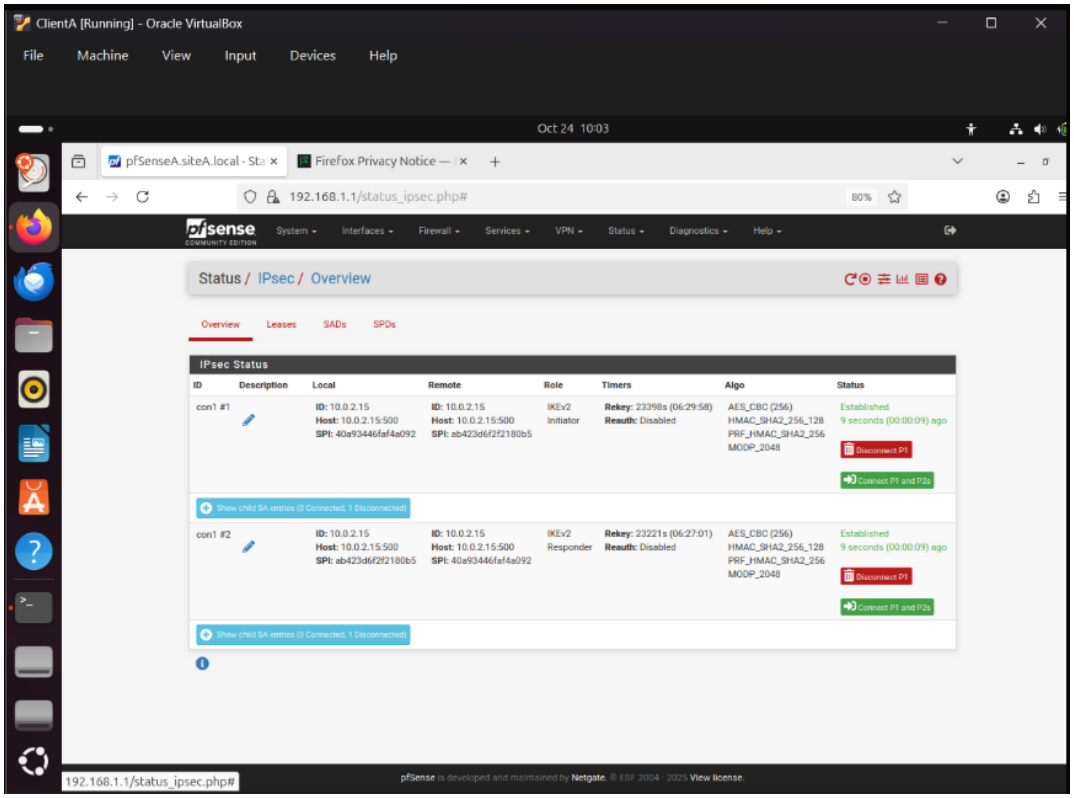


Figure 12

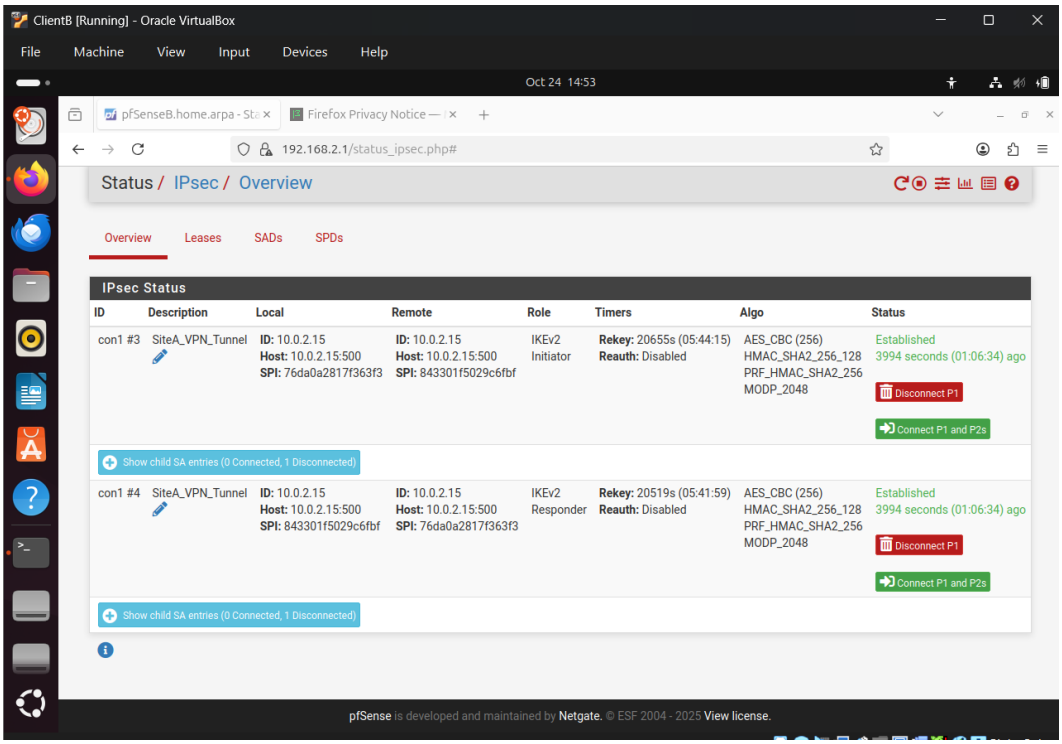


Figure 2: B Result

Conclusion

Using AES-256/SHA-256 encryption, the Site-to-Site VPN built with pfSense demonstrated communication between two LANs that was secure and confidential. Tunnel status and packet capture were used to confirm each step.
IKEv2 resilience and the significance of aligning Phase 1/2 policies were confirmed by the configuration. Secure routing was confirmed after firewall rules were changed to allow only ESP traffic. Ping tests for data flow revealed connectivity, and WAN captures verified payload encryption.
Using certificate-based authentication and implementing IDS features to keep an eye on IPsec traffic irregularities are potential future enhancements. All things considered, this lab achieved all of its goals and gave participants a practical grasp of encryption, VPN design, and enterprise-level secure routing.

Appendix – Configuration Files and Security Policies

A. Network Configuration Overview

Device	Role	WAN IP	LAN IP	Subnet	Gateway	Client IP
pfSense A	Site A Firewall	10.0.2.15	192.168.1.1	/24	10.0.2.2	192.168.1.101
pfSense B	Site B Firewall	10.0.2.4	192.168.2.1	/24	10.0.2.2	192.168.2.101

Purpose: To enable secure communication between Site A and Site B using an IPsec Site-to-Site VPN tunnel in pfSense.

Refer to **Figure 1 in Main Report** for VirtualBox VM overview.

B. Client Device IP Configuration Verification

Client A (Ubuntu)

Commands used:

```
ip addr show
ip route show
```

IP: 192.168.1.101
Gateway: 192.168.1.1

Client B (Ubuntu)

Commands used:

```
ip addr show
ip route show
```

IP: 192.168.2.101
Gateway: 192.168.2.1

Refer to **Figures 2 and 3** in the main report for screenshots of IP and routing tables.

C. pfSense Interface Configuration (WAN/LAN)

pfSense A

```
WAN (em0): 10.0.2.15/24
LAN (em1): 192.168.1.1/24
Gateway: 10.0.2.2
```

pfSense B

```
WAN (em0): 10.0.2.4/24
LAN (em1): 192.168.2.1/24
Gateway: 10.0.2.2
```

Refer to **Figures 4 and 5** for WAN/LAN interface status.

D. Phase 1 Configuration (IPsec Tunnel)

pfSense A

```
Key Exchange Version: IKEv2
Remote Gateway: 10.0.2.4
Authentication Method: Pre-Shared Key
```

```
Pre-Shared Key: vpn@123
Encryption Algorithm: AES 256
Hash Algorithm: SHA256
DH Group: 14 (2048-bit)
Lifetime: 28800 seconds
```

pfSense B

```
Key Exchange Version: IKEv2
Remote Gateway: 10.0.2.15
Authentication Method: Pre-Shared Key
Pre-Shared Key: vpn@123
Encryption Algorithm: AES 256
Hash Algorithm: SHA256
DH Group: 14 (2048-bit)
Lifetime: 28800 seconds
```

Refer to **Figures 6 and 7** for Phase 1 configuration screenshots.

E. Phase 2 Configuration

pfSense A

```
Local Network: 192.168.1.0/24
Remote Network: 192.168.2.0/24
Protocol: ESP
Encryption: AES (256-bit)
Hash: SHA256
PFS Key Group: 14
Lifetime: 3600 seconds
```

pfSense B

```
Local Network: 192.168.2.0/24
Remote Network: 192.168.1.0/24
Protocol: ESP
Encryption: AES (256-bit)
Hash: SHA256
PFS Key Group: 14
Lifetime: 3600 seconds
```

Refer to **Figures 8 and 9** in the main report for detailed Phase 2 setup.

F. Firewall Rules Configuration

pfSense A

```
Action: Pass
Interface: IPsec
Protocol: Any
Source: 192.168.1.0/24
Destination: 192.168.2.0/24
Description: Allow IPsec Site-to-Site Traffic
```

pfSense B

```
Action: Pass
Interface: IPsec
Protocol: Any
Source: 192.168.2.0/24
Destination: 192.168.1.0/24
Description: Allow IPsec Site-to-Site Traffic
```

Refer to **Figures 10 and 11** for screenshots of firewall rule configurations.

G. Security Policies Implemented

Security Policy	Purpose	Confirmation
AES-256 Encryption	Ensures strong confidentiality for transmitted data	Verified under IPsec SAD table
SHA-256 Hashing	Provides message integrity	Confirmed in IPsec logs
IKEv2 Protocol	Enables efficient key exchange and rekeying	Used in Phase 1 setup
DH Group 14	Adds Perfect Forward Secrecy	Confirmed in Phase 1 parameters
Pre-Shared Key	Authenticates both endpoints	vpn@123 used on both pfSenses
IPsec Firewall Rule	Restricts communication to encrypted traffic only	Rules visible under IPsec tab

Refer to **Figure 12** for IPsec tunnel status showing encryption details.

H. Encrypted Data Transfer Demonstration

Client A → Client B

```
ping 192.168.2.101
```

Output:

```
64 bytes from 192.168.2.101: icmp_seq=1 ttl=64 time=0.431 ms
3 packets transmitted, 3 received, 0% packet loss
```

Client B → Client A

```
ping 192.168.1.101
```

Output:

```
64 bytes from 192.168.1.101: icmp_seq=1 ttl=64 time=0.452 ms
3 packets transmitted, 3 received, 0% packet loss
```

Refer to **Figures 13 and 14** for screenshots of encrypted data transfer (ping results).

I. Troubleshooting Summary

Issue	Cause	Resolution
Tunnel stuck on "Connecting"	Incorrect WAN network	Changed both WANs to same Internal Network (WAN-NET)
No ping response	Missing IPsec rule	Added IPsec pass rule under Firewall tab
Phase 2 failure	Subnet mismatch	Corrected local/remote subnet entries
Authentication error	PSK mismatch	Re-entered vpn@123 on both ends

---

J. Validation & Final Evidence Checklist

Proof	Description
Figure 1	VirtualBox showing all VMs running
Figures 2–3	Client IP and Gateway confirmations
Figures 4–5	pfSense Interfaces (WAN/LAN)
Figures 6–7	Phase 1 Configuration (A & B)
Figures 8–9	Phase 2 Configuration (A & B)

Proof	Description
Figures 10–11	Firewall Rules (A & B)
Figure 12	IPsec Tunnel Established (Encryption Confirmation)
Figures 13–14	Ping Demonstration between Clients

K. Reflection on Security Design

This configuration ensures:

- All inter-site communication is encrypted with AES-256.
- Integrity and authentication are maintained with SHA-256 and Pre-Shared Keys.
- Perfect Forward Secrecy (PFS) is achieved using DH Group 14.
- Firewall rules restrict access strictly to VPN-encapsulated traffic.

Thus, the VPN topology demonstrates a secure, encrypted, and validated site-to-site communication link using pfS